LISA Link Simulation: Waveform to Six Links and TDI

LISA-TF simulation notes

July 21, 2026

Abstract

This note documents the simulation part of the repository: a user-provided gravitational-wave strain waveform is placed on a science time grid, propagated through a realistic LISA orbit to form six one-way link responses, injected into `lisainstrument` noise, and converted to Michelson `XYZ` and orthogonal `AET` TDI products with `PyTDI`. The diagnostic figures come from the MBHB run in `diagnostics/mbhb_sixlinks`.

1 Scope

The reusable entry point is `scripts/lisa_link_simulation.py`. It exposes a small facade around three pieces of code:

- `scripts/six_link_generation.py`: waveform preparation, orbit file, GW link response, and `lisainstrument` HDF5 generation.
- `scripts/tdi_computation.py`: `XYZ` and `AET` TDI from the full instrument HDF5 and from the GW-only HDF5.
- `scripts/plotting_utils.py`: diagnostic plots.

The simulation is intentionally time-domain until TDI products are optionally frequency-cut. It writes both science data and provenance files, so later notebooks can use the saved CSV/HDF5 outputs without rerunning the expensive instrument stage.

2 Pipeline

`t, hplus, hcross` → science-grid waveform → LISA orbit HDF5 → six GW links → `lisainstrument` HDF5 → `PyTDI` `XYZ` → `AET`

Stage	Main routine	Main output
Waveform alignment	`prepare_science_waveform`	`inputs/waveform_aligned.npz`
Orbit generation	`create_orbit_file`	`inputs/orbits.h5`
GW link response	`compute_gw_links`	`outputs/six_links_gw_only.csv`, `outputs/gws.h5`
Instrument run	`run_lisa_instrument`	`outputs/lisa_instrument_output.h5`
Noisy TDI	`compute_tdi_from_instrument`	`outputs/tdi_michelson_XYZ_AET.csv`
GW-only TDI	`compute_tdi_from_gws`	`outputs/tdi_gw_only_XYZ_AET.csv`

Table 1: Simulation stages and durable outputs.

3 Configuration

The simulation is configured with `LisaLinkSimulationConfig`. The current diagnostic convention is:

- time frame: TCB seconds since the LISA epoch;
- sky frame: ICRS right ascension and declination;
- link labels: `ij` means receiver `i`, emitter `j`;
- orbit dataset: `tcb/ltt`;
- GW dataset: `tcb`;
- GW links: dimensionless fractional frequency, with `_hz` columns produced by multiplying by the optical carrier frequency;
- TDI unit: frequency, so noisy and GW-only TDI products are compared in Hz.

The burn-in segment is prepended before the science segment. It lets `lisainstrument` and `PyTDI` settle before the saved science data starts. The orbit is generated longer than the waveform, typically `orbit_duration_factor = 5`, so the science grid lies safely inside the orbit validity range without generating an unnecessarily long orbit file.

Listing 1: Minimal configuration pattern.

```
from scripts.lisa_link_simulation import LisaLinkSimulationConfig

config = LisaLinkSimulationConfig(
    dt=0.1,
    science_duration=150_000.0,
    burn_in_seconds=600.0,
    edge_taper_seconds=10_000.0,
    frequency_low_hz=1e-5,
    frequency_high_hz=1e-1,
    orbit_duration_factor=5.0,
    orbit_dataset="tcb/ltt",
    gw_dataset="tcb",
    tdi_generation=2,
    tdi_unit="frequency",
)
```

4 Waveform Placement

The input waveform is a strictly increasing time array and two strain arrays, `hplus` and `hcross`. If `center_waveform_peak=True`, the largest strain-amplitude sample is placed in the center of the science segment. Outside the waveform support, the strain is zero-padded. An optional cosine edge taper suppresses boundary discontinuities.

5 Six Link Generation

The six one-way responses are computed with `lisagwresponse.ReadStrain` on the LISA orbit time grid. The saved CSV includes both the dimensionless fractional-frequency link responses and Hz-converted columns:

$$y_{ij}^{\mathrm{Hz}}(t) = \nu_0 \, y_{ij}(t),$$

where $\nu_0 = 2.816 \times 10^{14}\,\mathrm{Hz}$ is the optical carrier frequency used by the instrument simulation.

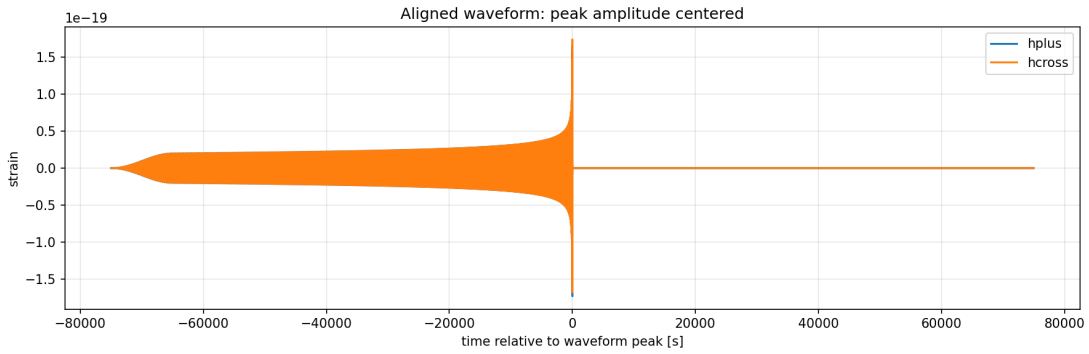


Figure 1: Prepared MBHB waveform after science-grid placement and edge taper.

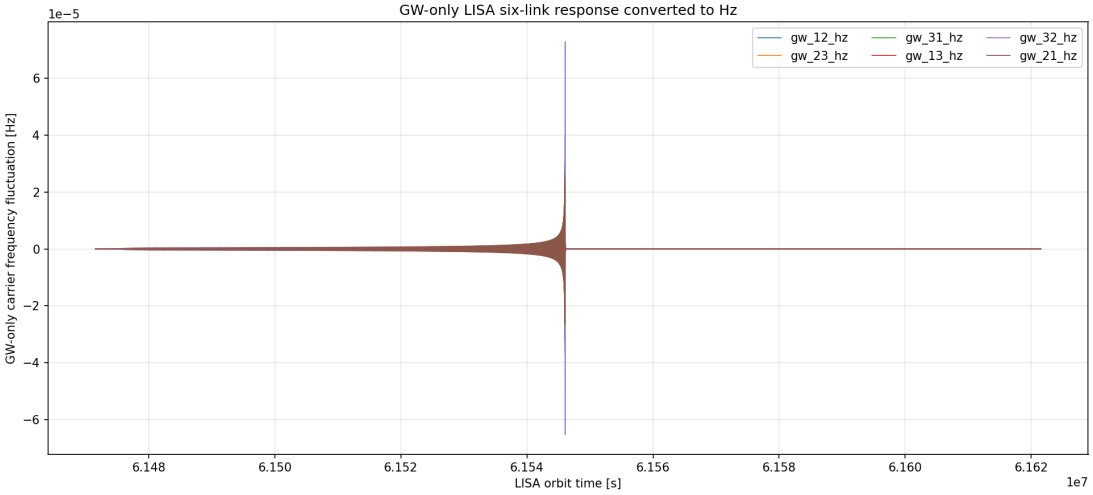


Figure 2: GW-only six-link response, converted to Hz for comparison with instrument outputs.

6 Realistic Instrument Noise

`lisainstrument.Instrument` reads the orbit HDF5 and the GW HDF5, adds the instrument noise model, and writes the full measurement HDF5. The exported `six_links_noisy.csv` is a lightweight science-window diagnostic from `debug/sci_carrier_fluctuations`; the realistic TDI products are computed from the full HDF5 through `PyTDI`, not only from these debug columns.

7 TDI Products

The noisy TDI path uses `Data.from_instrument(..., signals="fluctuations")` and evaluates Michelson `X2`, `Y2`, and `Z2`. Orthogonal channels are then formed as

$$A = \frac{Z - X}{\sqrt{2}}, \qquad E = \frac{X - 2Y + Z}{\sqrt{6}}, \qquad T = \frac{X + Y + Z}{\sqrt{3}}.$$

The GW-only TDI path uses `Data.from_gws` and then multiplies by the same carrier frequency so that the GW-only and noisy TDI products have consistent units.

The optional frequency cut is applied after TDI in the FFT domain. In the MBHB diagnostic shown here, the band is 10^{-5} to 10^{-1} Hz.

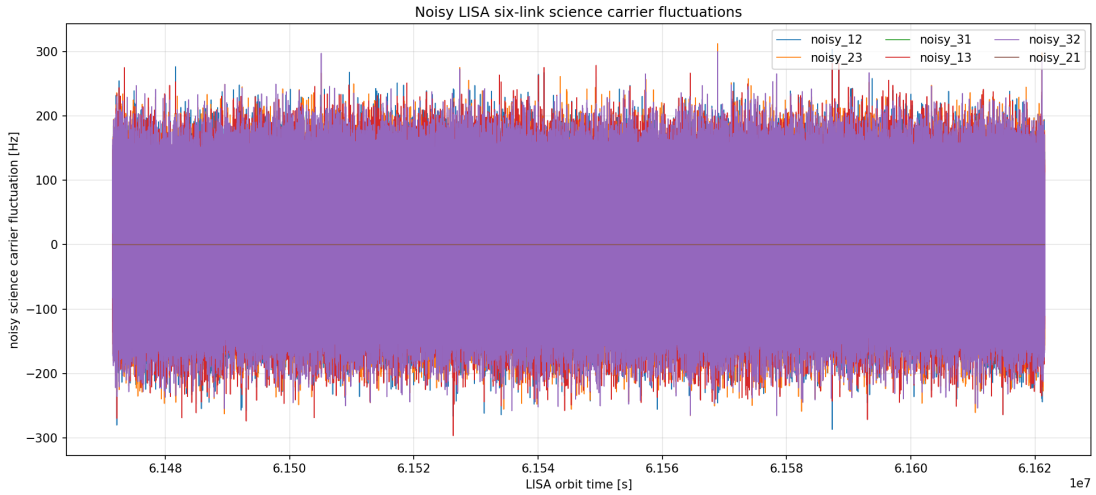


Figure 3: Science-window noisy link diagnostic from the instrument HDF5.

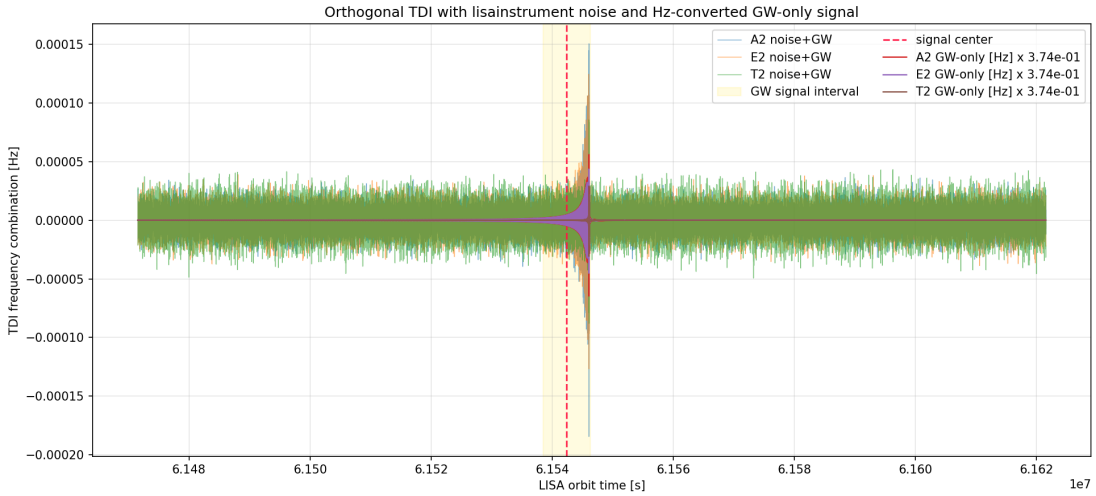


Figure 4: Noisy `AET` TDI with the Hz-converted GW-only signal overlaid for localization in time.

8 Public API

Listing 2: End-to-end simulation call.

```
from pathlib import Path
from scripts.lisa_link_simulation import (
    LisaLinkSimulationConfig,
    load_npz_waveform,
    run_lisa_link_simulation,
)

t, hplus, hcross = load_npz_waveform("lisa_link_sim/1e5-wave/inputs/
    my_waveform_1e5.npz")

products = run_lisa_link_simulation(
    t,
    hplus,
    hcross,
```

```
    work_dir=Path("examples/runs/minimal_lisa_link_simulation"),
    config=LisaLinkSimulationConfig(),
    make_plots=True,
)
```

The returned object contains `products.paths`, a structured list of all output paths, and `products.diagnostics`, scalar checks such as waveform duration, orbit span, burn-in size, TDI sample count, and signal timing.

9 SNR Convention

The SNR diagnostic follows Babak, Hewitson, and Petiteau, *LISA Sensitivity and SNR Calculations*, LISA-LCST-SGS-TN-001, arXiv:2108.01167. For a deterministic detector-output/TDI signal h,

$$\rho_{\mathrm{opt}} = \sqrt{(h|h)}, \qquad (a|b) = 4\,\mathrm{Re}\int_0^{f_{\max}} \frac{\tilde{a}^*(f)\tilde{b}(f)}{S_n(f)}\,df.$$

Here $S_n(f)$ is the one-sided PSD of the same TDI channel and in the same units as the signal. Explicit channels are combined by summing ρ_i^2.

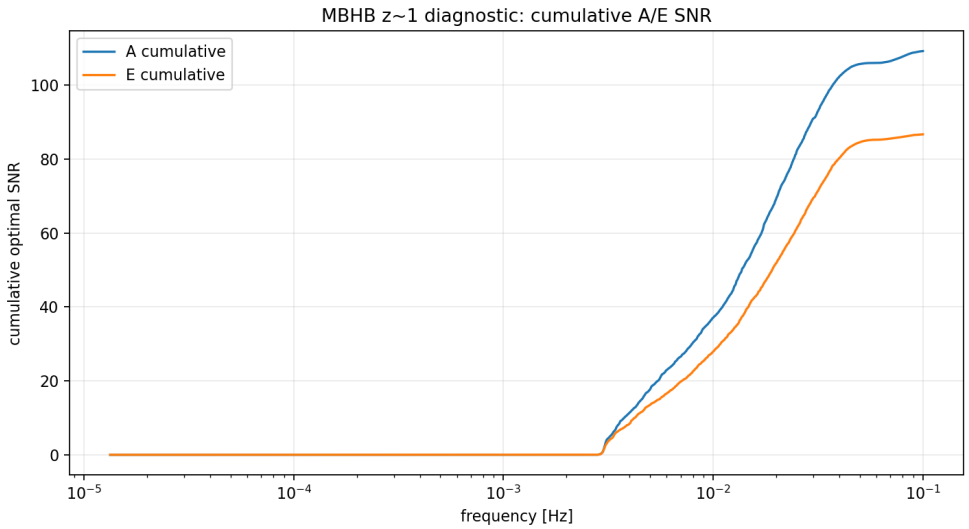


Figure 5: Cumulative `A/E` optimal SNR for the MBHB diagnostic using the one-sided TDI-channel PSD convention.

10 Minimal Output Contract

A completed simulation run writes:

- `six_links_gw_only.csv`: six GW link responses, dimensionless and Hz-converted;
- `six_links_noisy.csv`: science-window noisy link diagnostic;
- `tdi_michelson_XYZ_AET.csv`: noisy `XYZ/AET` TDI in Hz;
- `tdi_gw_only_XYZ_AET.csv`: GW-only `XYZ/AET` TDI in Hz;
- `orbits.h5`, `gws.h5`, and `lisa_instrument_output.h5`: durable HDF5 provenance.